

PRACTICAL 5

PRACTICAL TITLE :- Registration Form with Frontend Validation and User-Friendly Error Handling

Problem Definition	Create a student registration form with HTML5 input types and JavaScript validation for name, email, mobile number, password, confirm password, course, year, gender, and terms acceptance. Use Regular Expression for validations.
Key Questions / Analysis / Interpretation	1. Are correct input types and attributes used? 2. Are validation errors displayed near the relevant fields? 3. Is password strength checked? 4. Is the form accessible using keyboard and screen-reader-friendly labels?
Key Skills to be Addressed	HTML forms, JavaScript validation, regular expressions, password strength meter, UX error messages
Applications	Registration and data-entry systems
Learning Outcome	Students will design accessible forms with robust client-side validation.
CO Mapping	CO1, CO3
Dataset/Test Data	Valid and invalid sample registration inputs
Tools/Technology to be Used	HTML5, JavaScript ES5/ES6, browser dev tools
Total Hours	Implementation: 4 hours Total engagement: 6 hours
Post Laboratory Work Description	Submit validation test cases with screenshots for valid and invalid inputs.
Evaluation Strategy Including Viva	Input coverage, validation accuracy, error handling, and code walkthrough.
Advanced/Intermediate Extension	Intermediate: Add real-time validation on keyup/change events. Advanced: Add custom CAPTCHA using Canvas.

GIT HUB REPOSITORY :-

<https://github.com/nehalvala0005-cloud/sh.git>

1. Aim

To design and implement a student registration form using HTML5 input types and JavaScript validation with Regular Expressions, password strength checking, and user-friendly error messages.

2. Problem Definition

Create a student registration form using HTML5 input types and JavaScript validation for student details such as name, email, mobile number, password, confirm password, course, gender, and terms acceptance. Regular Expressions are used for validating appropriate input fields.

3. Key Questions / Analysis

Write these exactly:

1. Are correct HTML5 input types and attributes used?
 2. Are validation errors displayed near the relevant fields?
 3. Is password strength checked?
 4. Is the form accessible using keyboard and screen-reader-friendly labels?
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4. Key Skills to be Addressed

- HTML5 Forms
 - JavaScript Form Validation
 - Regular Expressions
 - Password Strength Checking
 - Error Message Handling
 - User-Friendly Form Design
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5. Application

Student registration and data-entry systems.

6. Theory

HTML5 Form

HTML5 provides different input types such as text, email, tel, password, and date to collect different types of information from users.

JavaScript Validation

JavaScript is used to validate the entered data before submitting the form. It checks whether the user has entered valid information and displays appropriate error messages.

Regular Expression

Regular Expressions are patterns used to validate specific formats.

For example:

- Email → valid email format
- Mobile → 10-digit number
- Name → alphabets and spaces
- Password → uppercase, lowercase and number

Password Strength

The password is checked for minimum length and required character types. The strength message helps the user create a stronger password.

7. Implementation

Yaha apne actual register.html aur register.js code add karna.

Agar practical file me pura code nahi maanga hai, to important JavaScript validation code ka portion bhi enough hai.

Important validations implemented:

- Name validation
- Email validation
- Mobile number validation
- Password validation
- Confirm password matching
- Gender selection
- Course selection

- Terms and conditions acceptance
- Password strength indication
- Error messages near fields

8. Test Cases

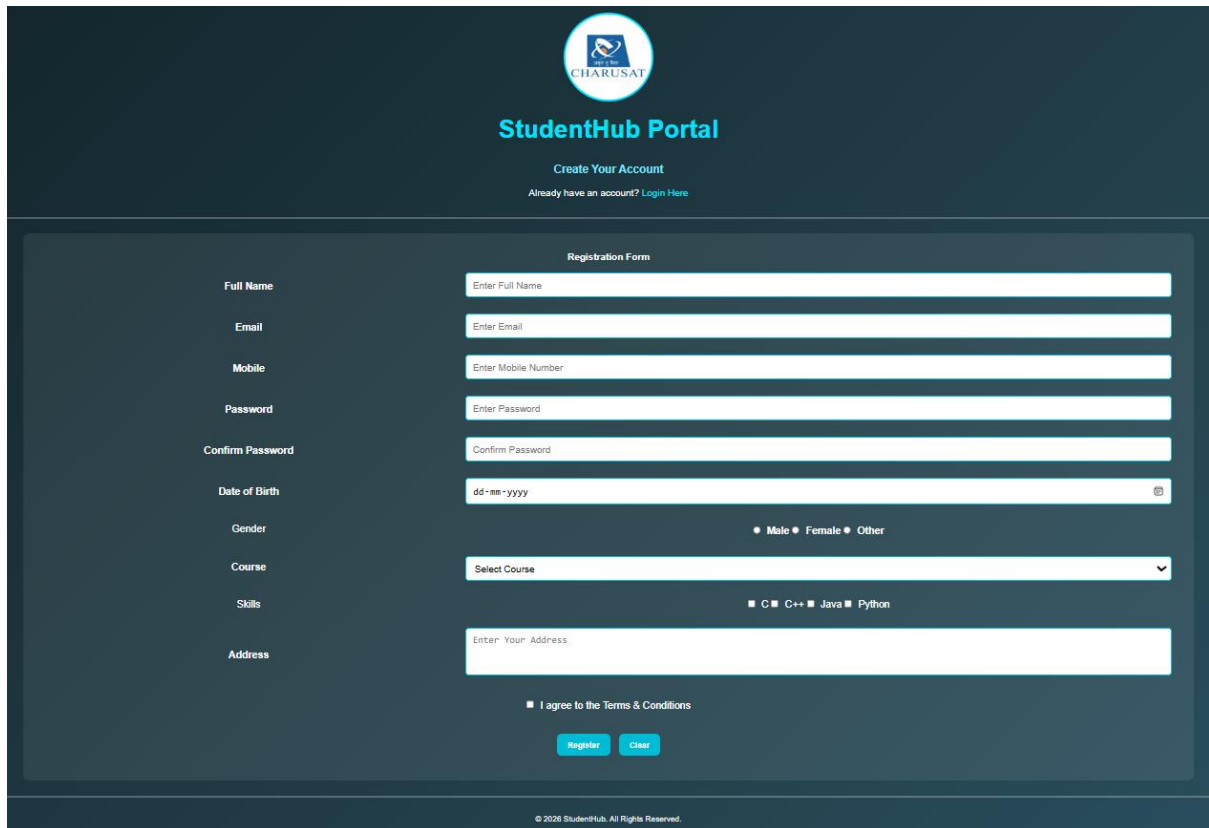
Test Case Input		Expected Result
1	Valid name	Accepted
2	Invalid email	Error displayed
3	Invalid mobile number	Error displayed
4	Weak password	Password strength/error displayed
5	Different confirm password	Error displayed
6	Course not selected	Error displayed
7	Gender not selected	Error displayed
8	Terms not accepted	Error displayed
9	All valid inputs	Registration successful

9. Result

The student registration form was successfully implemented using HTML5 and JavaScript. Client-side validation was performed using Regular Expressions, password strength checking, and appropriate error messages.

PROOF OF WORK :-

5.1 :- Registration Form



The screenshot displays the 'StudentHub Portal' registration interface. At the top, the CHARUSAT logo is visible, followed by the text 'StudentHub Portal', 'Create Your Account', and a link 'Already have an account? Login Here'. The main registration form is titled 'Registration Form' and includes the following fields and options:

- Full Name:** Text input field with placeholder 'Enter Full Name'.
- Email:** Text input field with placeholder 'Enter Email'.
- Mobile:** Text input field with placeholder 'Enter Mobile Number'.
- Password:** Text input field with placeholder 'Enter Password'.
- Confirm Password:** Text input field with placeholder 'Confirm Password'.
- Date of Birth:** Text input field with placeholder 'dd-mm-yyyy' and a calendar icon.
- Gender:** Radio button options: Male, Female, Other.
- Course:** Dropdown menu with placeholder 'Select Course'.
- Skills:** Checkable options: C, C++, Java, Python.
- Address:** Text input field with placeholder 'Enter Your Address'.

Below the form fields, there is a checkbox for 'I agree to the Terms & Conditions'. At the bottom of the form are two buttons: 'Register' and 'Clear'. The footer of the page states '© 2026 StudentHub. All Rights Reserved.'.

5.2 :- Invalid Input Validation

StudentHub Portal

Create Your Account

Already have an account? [Login Here](#)

Registration Form

Full Name	Nehal Mukeshbhai Vala
Email	NEHAL452GMAIL.COM Enter a valid email address.
Mobile	NEH1235 Enter exactly 10 digits.
Password	***** Weak password Password must contain at least 8 characters.
Confirm Password	Confirm Password Please confirm your password.
Date of Birth	08-12-2007
Gender	☐ Male ☐ Female ☐ Other
Course	Select Course

5.3 :- Password Strength

Weak password
Password must contain at least 8 characters.

Medium password
Use uppercase, lowercase and a number.

Strong password

5.4 :- Valid Registration

The screenshot displays the 'StudentHub Portal' registration page. A central white popup with a green checkmark and the text 'Congratulations! You are registered successfully.' is overlaid on the form. The form fields include: Full Name, Email, Mobile, Password, Confirm Password, Date of Birth (set to 08-12-2007), Gender (radio buttons for Male, Female, Other), and Course (a dropdown menu currently showing 'B.Tech CE'). The background is a dark blue gradient with the portal's logo and navigation links.

5.5 :- Validation testing of the student registration form

This screenshot shows the same registration form as in 5.4, but with the browser's developer tools open. The 'Elements' panel on the right shows the DOM structure, highlighting the success popup with the class 'success-popup'. The 'Styles' panel shows the computed styles for the popup, including its position and background. The 'Console' panel at the bottom displays two error messages: 'The AudioContext was not allowed to start. It must be resumed (or created) after a user gesture on the page.' and 'Failed to load resource: the server responded with a status of 404 (Not Found)'. The browser's address bar shows the URL '127.0.0.1:5500/StudentHub/web/Html/register.html?gender=on'.

Learning Outcomes

After completing this practical, I was able to:

- Understand and use HTML5 form elements and input types.
- Implement client-side form validation using JavaScript.
- Use Regular Expressions to validate user inputs.
- Implement password strength checking.
- Display meaningful error messages near the relevant fields.
- Understand the importance of accessible and user-friendly forms.
- Test valid and invalid inputs using different validation test cases.

Conclusion

- Thus, the student registration form was successfully designed using HTML5 and JavaScript. Different input fields were validated using JavaScript and Regular Expressions, while password strength and confirmation were also checked. Appropriate error messages were displayed for invalid inputs, making the form more user-friendly and reliable. This practical helped in understanding client-side validation and the development of accessible web forms.

Faculty Remarks :- _____